THE CO-OPERATIVE UNIVERSITY OF KENYA (CUK)

SCHOOL OF COMPUTING AND MATHEMATICS

LOAN DEFAULT PREDICTION USING CLASSIFICATION TECHNIQUES.

BY

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Research submitted in partial fulfillment of the requirements for the award of the Bachelor of Science in Computer Science.

December, 2024.

DECLARATION AND APPROVAL DECLARATION

DECLARATION

I hereby declare that this Project Loan Default Prediction System Using Classification Techniques is my own work and has, to the best of my knowledge, not been submitted to any other institution of higher learning for any award.

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APPROVAL

This project Loan Default Prediction System Using Classification Techniques has been submitted with my approval as the University supervisor.

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ABSTRACT

This project explores the development of a machine learning-based model to predict loan defaults, aimed at assisting financial institutions in evaluating applicant risk more accurately. Loan defaults present significant challenges to financial institutions, impacting their profitability and risk management strategies. Traditional risk assessment methods may overlook subtle patterns in data, whereas machine learning models can leverage complex relationships within applicant data to make reliable predictions. This project evaluates several classification models—including Logistic Regression, Decision Tree, Random Forest, and Neural Networks—using financial and demographic features such as income, credit score, and debt-to-income ratio. The chosen model is deployed through a REST API for integration with a mobile application, enabling real-time access for loan officers. The proposed system enhances decision-making, reduces the likelihood of defaults, and streamlines loan processing.

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LIST OF ABBREAVIATIONS

Abbreviation	Definition
ML	Machine Learning
SVM	Support Vector Machine
GBM	Gradient Boosting Machine
API	Application Programming Interface
AUC-ROC	Area Under the Curve – Receiver Operating Curve
SQL	Structured Query Language
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
CRISP-DM	Cross-Industry Standard Process for Data Mining
AWS	Amazon web Services

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CHAPTER 1: INTRODUCTION

1.1 Background to the Study

Financial institutions faced substantial challenges in assessing loan applications accurately due to potential defaults. Loan defaults disrupted profitability and affected the institution's risk portfolio, resulting in heightened scrutiny in lending decisions. Conventional assessment methods relied on manual reviews or rule-based criteria, which often failed to identify complex data patterns associated with loan default risks. Leveraging machine learning, this project proposed a predictive model that classified applicants based on historical data, empowering financial institutions to make data-driven decisions, reduce defaults, and improve operational efficiency.

1.2 Statement of the Problem

Loan defaults presented a significant financial risk to lending institutions, affecting their asset quality and profitability. The existing methods of applicant risk assessment were often manual, rule-based, and insufficient for identifying subtle patterns in applicant data. Consequently, this project aimed to develop a machine learning model to classify applicants based on default risk accurately. Such a model offered an automated, data-driven approach to improve loan approval processes and reduce potential financial losses.

1.3 Objectives

1.3.1 General Objectives

To develop a machine learning model for predicting loan default risk based on applicant data, enabling financial institutions to make informed lending decisions and reduce financial losses.

1.3.2 Specific Objectives

1. To collect and preprocess loan applicant data, including financial and demographic attributes.

2. To train multiple classification models, such as Logistic Regression, Decision Tree, Random Forest, GBM, XGBoost, SVM, and Neural Networks, and evaluate them on accuracy, precision, recall, F1 score, and AUC-ROC metrics.
3. To perform hyperparameter tuning to optimize model performance.
4. To analyze feature importance to identify key predictors of loan default.
5. To deploy the final model via a REST API, enabling integration with a cross-platform mobile application.

1.4 Purpose of the Study

The purpose of this study was to develop an automated, predictive loan default assessment model that helped financial institutions minimize default risks and improve decision-making. The system aimed to streamline the risk assessment process, enabling scalability and efficiency in loan processing.

1.5 Significance of the Study

The proposed loan default prediction system benefited financial institutions by:

- Improving Risk Assessment: Identifying high-risk applicants for enhanced loan portfolio management.
- Enabling Data-Driven Decisions: Replacing manual assessments with automated, accurate predictions.
- Enhancing Operational Efficiency: Enabling faster loan approvals and scalability in loan processing.

1.6 Scope of the Study

This study focused on building a predictive loan default model using machine learning classification techniques. Data required for the project included demographic and financial information on applicants, and the study emphasized model deployment as an API integrated with a mobile app for real-time use by loan officers.

1.7 Assumptions

1. The dataset represented a wide range of loan applicants, capturing diverse profiles.
2. Machine learning techniques and evaluation metrics used were suitable for loan default prediction.
3. The mobile application enabled effective interaction with the deployed prediction model.

1.8 Limitations and Delimitations

1.8.1 Limitations

1. Data Availability and Quality: Data quality could have impacted model accuracy.
2. Computational Resources: Advanced models required significant resources for training.
3. Model Generalization: The model needed further adaptation for different loan types or applicant profiles.

1.8.2 Delimitations

1. This study focused on supervised classification techniques, excluding unsupervised learning approaches.
2. Deployment was limited to a REST API for mobile app integration, without additional web or desktop interfaces.

1.9 Definition of Terms

Loan Default: Failure of a borrower to fulfill loan payment obligations.

Machine Learning: Algorithm-driven data analysis that enabled pattern recognition for predictions.

Classification Model: A model that categorized data into classes—in this case, default or non-default.

Hyperparameter Tuning: Adjusting model parameters to enhance predictive performance.

Feature Importance: Analysis of features that significantly contributed to model predictions.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Loan default prediction played a crucial role in helping financial institutions minimize risks and losses. By identifying borrowers likely to default, institutions made better lending decisions and improved overall portfolio performance. Despite the progress made through traditional credit scoring systems and machine learning-based tools, existing solutions often failed to adapt to emerging challenges. This review explored existing methods, highlighted their weaknesses, and discussed how the proposed system aimed to address these gaps with a fresh and innovative approach.

2.2 Loan Default Prediction

The goal of loan default prediction was to use a borrower's financial and personal data to estimate their likelihood of repaying a loan. Traditional systems, such as credit scores, relied on predefined metrics, offering broad evaluations but often missing nuanced patterns. Machine learning transformed this field by using historical data to detect complex relationships between variables. However, even advanced machine learning models faced challenges in practical applications, such as transparency, scalability, and adaptability.

2.3 Existing Systems

Several systems had been implemented to assess loan default risk. These systems demonstrated significant advancements, yet they also had limitations that impacted their effectiveness.

2.3.1 Credit Scoring Models (e.g., FICO Scores)

These were rule-based systems that evaluated borrowers based on fixed parameters, such as repayment history and credit utilization. While reliable for basic risk assessments, they lacked the flexibility to incorporate non-traditional data sources or detect subtle patterns in borrower behavior.

2.3.2 AI-based Loan Risk Platforms

Platforms such as Zest AI and other proprietary tools used by financial institutions employed machine learning to predict risk. While they achieved high accuracy, these systems often functioned as black boxes, providing little insight into how predictions were made. Additionally, their focus on larger datasets made them less accessible to smaller financial institutions.

2.3.3 Custom In-house Models

Many banks developed their own prediction systems tailored to their operations. While these models aligned closely with the institutions' needs, they were resource-intensive to develop and maintain. Moreover, they were often designed for internal use, limiting their scalability.

2.4 Weaknesses of Existing Systems

Existing systems faced several common challenges:

1. Static Models: Many systems relied on static data, making them unable to adapt to changing borrower behavior or economic conditions in real time.
2. Limited Features: The scope of features used was often narrow, excluding critical contextual or behavioral indicators.
3. Lack of Transparency: Advanced models, particularly neural networks, lacked interpretability.
4. Scalability Issues: Sophisticated systems required significant computational resources.
5. Imbalanced Datasets: Loan datasets often had a class imbalance, skewing predictions and reducing model reliability.

2.5 How the Proposed System Addressed These Weaknesses

The proposed system aimed to overcome the limitations of existing systems with a unique and practical approach:

1. **Dynamic Data Integration**

- Weakness Addressed: Static models.
- Solution: The system incorporated real-time data streams, allowing it to adapt to evolving borrower behaviors and market conditions.

2. **Broader Feature Scope**

- Weakness Addressed: Limited features.
- Solution: Advanced exploratory data analysis identified and included unconventional data points for a more comprehensive risk assessment.

3. **Explainable Predictions**

- Weakness Addressed: Lack of transparency.
- Solution: The system used tools like SHAP to explain model outputs, ensuring transparency and trust.

4. **Cloud-based Scalability**

- Weakness Addressed: Scalability issues.
- Solution: Hosting the system on cloud platforms like AWS or GCP provided a cost-effective, scalable solution.

5. **Balanced Predictions with SMOTE**

- Weakness Addressed: Imbalanced datasets.
- Solution: Techniques like SMOTE were used to address class imbalance, improving performance on default cases.

6. **Cost-effective Accessibility**

- Weakness Addressed: High resource demands.
- Solution: A lightweight and modular architecture ensured easy deployment even for smaller institutions.

2.6 Uniqueness of the Proposed Solution

The proposed system offered a unique value proposition:

- Behavioral-driven Predictions
- Hybrid Modeling Approach
- Transparency and Trust
- Localized Insights
- Inclusivity for institutions of all sizes

2.7 Conclusion

While existing loan default prediction systems had advanced significantly, their limitations in adaptability, transparency, and scalability left room for improvement. The proposed system addressed these gaps with innovative methods, ensuring it was dynamic, interpretable, and accessible.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter outlined the methodology for building the Loan Default Prediction System. The CRISP-DM framework guided the project. Each step ensured that the final system met the desired goals of accuracy and reliability. This approach balanced structure with flexibility, allowing adjustments at any stage to improve outcomes.

3.2 Project Design

3.2.1 Development Methodology

The project adopted the CRISP-DM framework. The six phases were:

- Business Understanding: Defined the problem and success criteria.
- Data Understanding: Explored datasets to identify patterns and inconsistencies.
- Data Preparation: Cleaned and organized data for modeling.
- Modeling: Built multiple machine learning models.
- Evaluation: Evaluated models using defined metrics.
- Deployment: Integrated the final model into a prediction system.

3.2.2 Justification for CRISP-DM

The CRISP-DM framework ensured a strong focus on data preparation and iterative improvements. Its flexibility supported continuous alignment with business needs.

3.3 Design Procedures

3.3.1 Requirement Analysis

A checklist of system requirements was created, answering questions like:

- What data would be processed?
- What should the output be?
- What constraints existed?

3.3.2 System Architecture

The system followed a client-server architecture. The server handled data processing and predictions, while the client interface allowed user interaction.

3.3.3 Workflow Design

The workflow was modular:

- Data cleaning through preprocessing module
- Separate model training
- Evaluation using metrics

3.4 System Requirements

3.4.1 Functional Requirements

The system:

- Imported raw datasets
- Cleaned and transformed data
- Trained and evaluated models
- Predicted loan default probabilities
- Provided secure API access
- Logged predictions for auditing

3.4.2 Non-Functional Requirements

- Security via HTTPS
- 99.9% reliability
- Scalability for growing user base
- Real-time performance (<1 second predictions)

3.4.3 Software Requirements Included:

- Python
- Libraries: pandas, NumPy, scikit-learn, XGBoost
- PostgreSQL

- Seaborn, Matplotlib, Plotly
- Jupyter, Anaconda, VS Code

3.4.4 Hardware Requirements Development:

- Intel i7 or equivalent
- 8GB+ RAM
- SSD storage Deployment:
- Cloud server (4+ CPUs, 16GB+ RAM)

3.5 Data Collection and Analysis

3.5.1 Data Collection Used:

- Public Datasets (Kaggle)

3.5.2 Data Analysis

- Univariate Analysis for trends
- Bivariate Analysis for relationships
- Correlation Analysis for redundancy

3.5.3 Data Preprocessing

- Handled missing data
- Normalized features
- Detected and addressed outliers

3.6 Model Training and Evaluation

3.6.1 Algorithms Used

- Logistic Regression
- Random Forest
- XGBoost

3.6.2 Evaluation Metrics

- Accuracy
- Precision, Recall
- F1 Score
- AUC-ROC

3.7 Deployment

The model was not deployed as a REST API hosted on AWS.

3.8 Ethical Considerations

- Data Privacy: Ensured anonymization
- Bias Mitigation: Used diverse datasets
- Explainability: Integrated SHAP for transparency

3.9 Conclusion

This methodology ensured a structured, iterative approach to solving loan default prediction. Each phase aligned technical decisions with business goals, resulting in a transparent, scalable, and reliable solution.

Chapter 4: System Analysis

4.1 Introduction

System analysis is a pivotal phase in the development of any predictive system, aimed at dissecting the current system's components, processes, and interactions to define the requirements for an enhanced solution. In the context of this project, titled "Loan Default Prediction," the system analysis focuses on evaluating the existing state of loan application data processing within financial institution and specifying the requirements for a machine learning-based predictive model to forecast loan default probabilities.

4.2 System Analysis

The current system involves a financial institution managing loan applications, capturing data such as applicant income, loan amount, credit history, and demographic details, but lacking a robust mechanism to predict loan defaults accurately. The absence of predictive analytics leads to suboptimal risk assessment, potentially resulting in higher default rates and financial losses. The proposed system introduces a machine learning-based predictive model to analyze historical loan data and predict the likelihood of default, enhancing decision-making for loan approvals.

4.2.1 Flow Charts

The flow chart outlines the sequential processes executed in the loan default prediction system:

1. **Data Collection**: Loading the loan dataset from a CSV file sourced from a reliable repository (Kaggle).
2. **Data Preprocessing**: Dropping irrelevant columns (e.g., 'Loan_ID'), handling missing values using median imputation, and encoding categorical variables with one-hot encoding.
3. **Exploratory Data Analysis (EDA)**: Visualizing the target variable distribution (e.g., 'Loan_Status') to understand class balance.

4. **Feature Engineering**: Creating new features such as 'TotalIncome', 'EMI', 'Debt_to_Income_Ratio', 'Loan_to_Income_Ratio', and 'Income_Stability' to enhance predictive power.
5. **Data Scaling**: Applying standardization to numerical features for model compatibility.
6. **Model Selection and Training**: Splitting data into training and testing sets, training a Random Forest Classifier within a pipeline that includes preprocessing.
7. **Model Evaluation**: Assessing performance using metrics like accuracy, precision, recall, F1 Score, and ROC-AUC.
8. **Deployment Preparation**: Saving the trained model and preprocessor for integration into a predictive application.

This sequence ensures a systematic progression from raw data to actionable predictions.

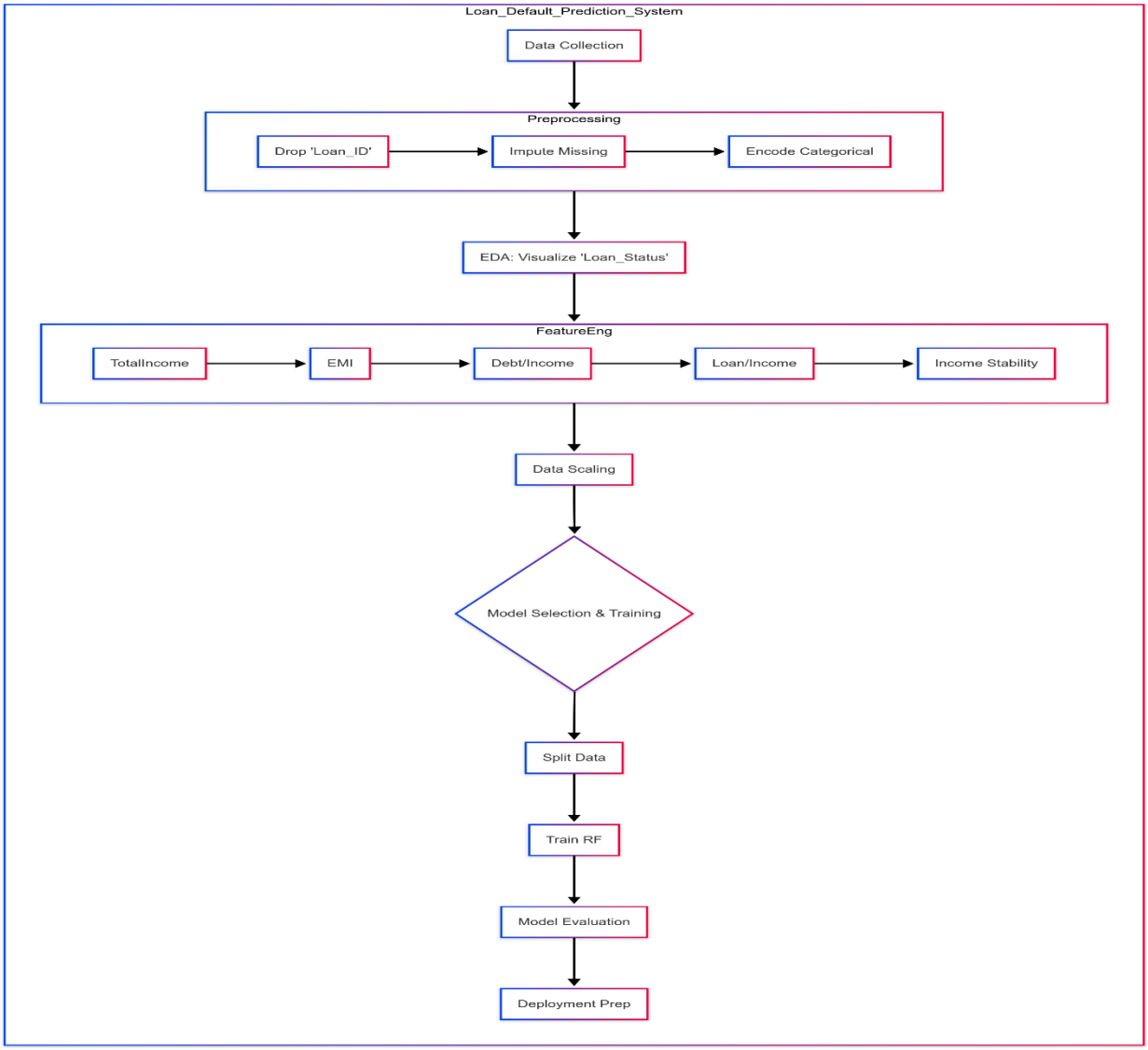

Figure 1 system flow chart

4.2.2 Data Flow Diagram (DFD)

The Data Flow Diagram illustrates the movement of data through the system:

- **Data Source**: The loan dataset (CSV file) containing raw applicant data.
- **Preprocessing Module**: Processes raw data by cleaning, encoding, and scaling it.
- **Feature Engineering Module**: Generates derived features from the preprocessed data.
- **EDA Module**: Analyzes data distributions and relationships (e.g., via count plots).

- **Modeling Module**: Trains the Random Forest model using the engineered features.
- **Evaluation Module**: Outputs performance metrics based on test data predictions.
- **Storage Module**: Saves the model and preprocessor (e.g., as 'loan_default_model.pkl' and 'preprocessor.pkl').
- **Prediction Interface**: Accepts user inputs, processes them through the saved model, and returns default probabilities.

This flow ensures data is transformed and utilized effectively for prediction.

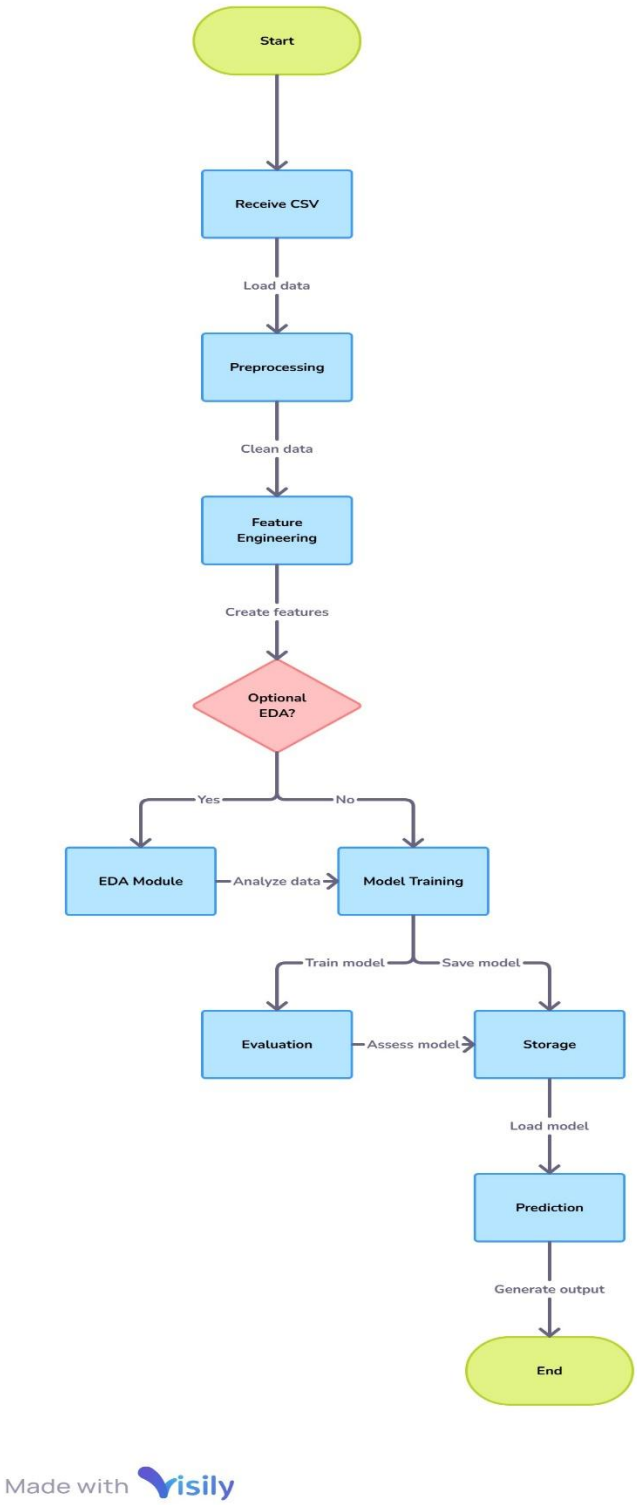

Figure 2 system data flow diagram

4.2.3 Class Diagram

Although the notebook is data-centric rather than object-oriented, a conceptual class diagram can be inferred:

- **DataLoader**: Manages loading of the CSV dataset.
  - Attributes: File path
  - Methods: Load_data()
- **Preprocessor**: Handles data cleaning and encoding.
  - Attributes: Imputer, Scaler, Encoder
  - Methods: Impute_missing(), Scale_features(), Encode_categorical()
- **FeatureEngineer**: Creates new features.
  - Attributes: Base features
  - Methods: Calculate_total_income(), Calculate_EMI()
- **ModelTrainer**: Trains the predictive model.
  - Attributes: Model (RandomForestClassifier), Train_data
  - Methods: Fit_model(), Evaluate_model()
- **Predictor**: Generates predictions.
  - Attributes: Trained_model
  - Methods: Predict_proba()

These classes interact to process data and produce predictions, with the Preprocessor feeding into FeatureEngineer, which supplies data to ModelTrainer, and finally to Predictor.

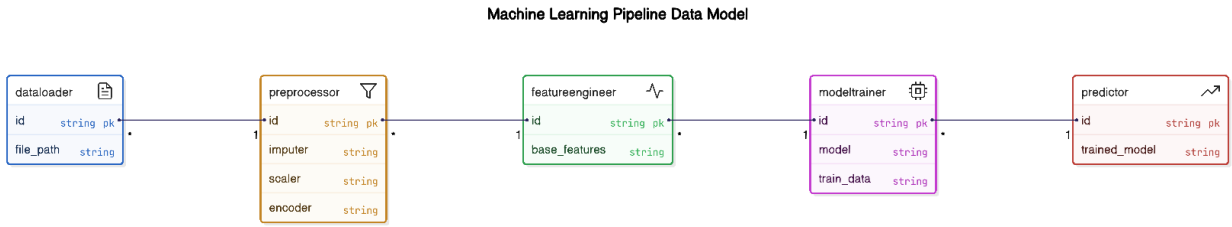

Figure 3 system class diagram

4.2.4 Use Case Diagram

The use case diagram captures user interactions with the system:

- **Actors**:
  - **Loan Officer**: Uses predictions to assess loan applications.
  - **Data Analyst**: Analyzes model performance and data insights.
- **Use Cases**:
  - **Load Data**: Data Analyst imports the dataset.
  - **Preprocess Data**: Data Analyst cleans and prepares data.
  - **Perform EDA**: Data Analyst visualizes data distributions.
  - **Train Model**: Data Analyst builds and evaluates the model.
  - **Make Prediction**: Loan Officer inputs applicant data to receive default probability.

Relationships show the Loan Officer relying on predictions, while the Data Analyst manages the backend processes.

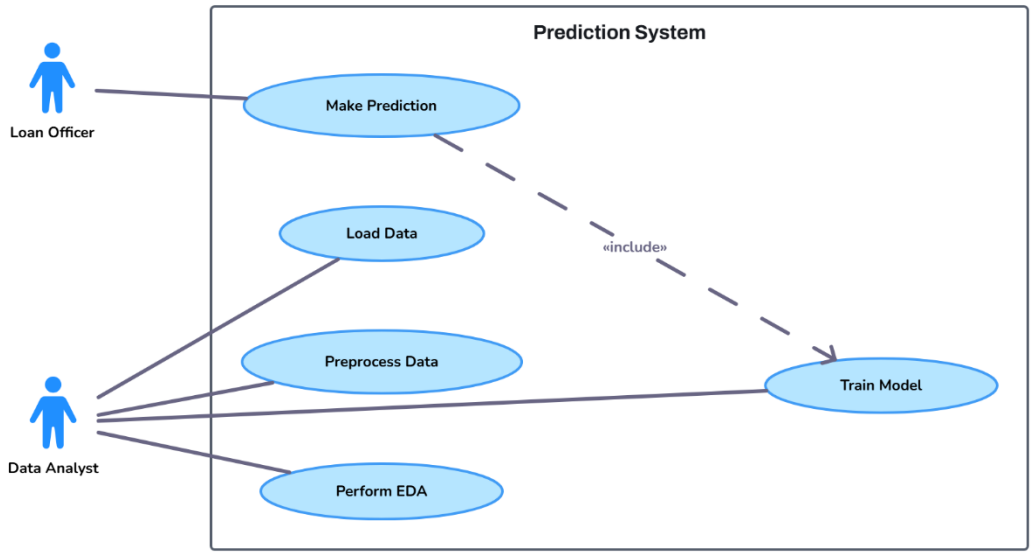

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Figure 4 system use case diagram

4.3 System Requirements

The system requirements are categorized into functional and non-functional aspects.

4.3.1 Functional Requirements

1. **Data Collection and Preprocessing**: The system must load loan data from a CSV file and preprocess it by handling missing values, encoding categorical variables, and scaling numerical features.
2. **Exploratory Data Analysis**: Provide visualization tools (e.g., count plots) to analyze the distribution of loan statuses and other variables.
3. **Feature Engineering**: Generate derived features like 'TotalIncome' and 'EMI' to improve model performance.
4. **Model Training and Evaluation**: Train a Random Forest Classifier and evaluate it using metrics such as precision, accuracy and ROC-AUC.
5. **Prediction Generation**: Enable predictions of loan default probabilities based on new applicant data.
6. **Model Persistence**: Save the trained model and preprocessor for future use.

4.3.2 Non-functional Requirements

1. **Performance**: The system should process data and generate predictions efficiently, handling datasets of at least 614 records (as per the dataset) within reasonable timeframes.
2. **Accuracy**: The model should achieve reliable performance, with metrics comparable to those observed (e.g., optimized via RandomizedSearchCV).
3. **Scalability**: Capable of processing larger datasets as the institution’s data grows.
4. **Usability**: The prediction function should accept raw JSON inputs and return clear outputs (e.g., prediction and probability).

5. **Reliability**: Ensure consistent model performance across different data splits, as achieved through stratified train-test splitting.

CHAPTER 5: SYSTEM DESIGN

Introduction

This chapter outlines the system design for the Loan Default Prediction System, which predicts the likelihood of loan defaults using a Kaggle dataset. The design encompasses architectural design, system overview, physical overview, data collection and preprocessing, and prediction mechanisms. It builds on the analysis from Chapter 4 and the machine learning workflow implemented in Python, ensuring a modular, efficient, and deployable solution for financial institutions.

5.1 System Overview

The Loan Default Prediction System processes applicant data to predict whether a loan is likely to default, aiding loan officers in risk assessment. It ingests raw data (e.g., income, credit history), preprocesses it (imputation, feature engineering, scaling, encoding), applies a trained machine learning model, and outputs a binary prediction (0 = No Default, 1 = Default) with a probability score. Predictions are stored for auditing and potential retraining, with monitoring for data drift to maintain model reliability. The system is designed for standalone use or integration via an API, leveraging a Kaggle dataset for training and evaluation.

Key Objectives:

- Accurate prediction of loan default risk.
- Efficient preprocessing of applicant data.
- Persistent storage of predictions for analysis.
- Monitoring to detect changes in data distribution.

5.2 Architectural Design

The system adopts a modular architecture with distinct components for data handling, processing, prediction, storage, and monitoring. It is designed to be lightweight, portable, and extensible for future API or server-based deployment.

5.2.1 Components

- **Data Ingestion Module**: Accepts applicant data as JSON or CSV, validating fields like ApplicantIncome, Gender, and LoanAmount.
- **Preprocessing Module**: Cleans data, engineers features (e.g., TotalIncome), and applies scaling/encoding using a Scikit-learn Pipeline.
- **Machine Learning Module**: Employs a RandomForestClassifier to generate predictions and probabilities.
- **Prediction Interface**: A Python function (potentially exposed as an API) that processes inputs and returns results.
- **Storage Module**: Uses SQLite to log predictions (applicant_id, prediction, actual_outcome, timestamp).
- **Monitoring Module**: Detects data drift with Evidently, generating HTML reports.

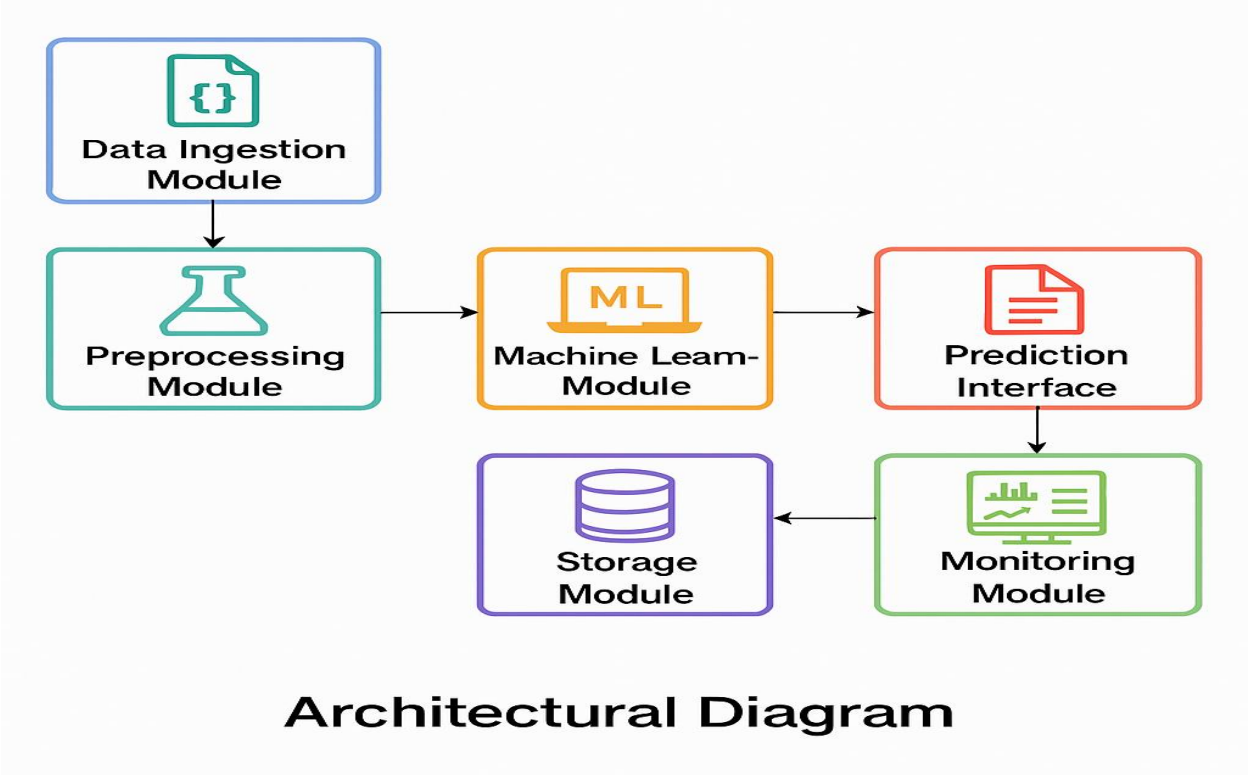


Figure 5 system architectural design

Description: Applicant data enters via the ingestion module, flows to preprocessing for feature engineering and transformation, and is processed by the ML module for prediction. The prediction interface returns results, which are stored in SQLite. The monitoring module periodically analyzes data drift.

5.2.2 Data Flow

1. Raw data (e.g., JSON with ApplicantIncome, Credit_History) is validated.
2. Preprocessing imputes missing values, computes features (e.g., EMI), scales numerical data, and encodes categorical variables.
3. The ML module predicts default risk using the trained model.
4. Results (prediction, probability) are returned and logged in SQLite.
5. Drift analysis compares new data to the training set, flagging distribution shifts.

5.3 Physical Overview

The system is designed for deployment on a single machine or containerized environment, with minimal hardware requirements suitable for development or small-scale production.

5.3.1 Hardware Requirements

- **Processor**: Multi-core CPU (e.g., Intel i5 or equivalent) for model training and inference.
- **Memory**: 8 GB RAM to handle dataset processing (Kaggle dataset ~614 rows).
- **Storage**: 10 GB disk space for dataset, model files, SQLite database, and logs.
- **Optional**: GPU not required, as the RandomForestClassifier runs efficiently on CPU.

5.3.2 Software Requirements

- **Operating System**: Windows, Linux, or macOS.
- **Python**: Version 3.12, with libraries:

    - Scikit-learn: For Pipeline, RandomForestClassifier, StandardScaler, OneHotEncoder.
    - Pandas: Data manipulation.
    - SQLite3: Database storage.
    - Evidently: Data drift monitoring.
    - Joblib: Model serialization.
    - FastAPI (proposed): For potential API deployment.
- **Development Tools**: Jupyter Notebook, optional IDE (e.g., VS Code).
- **Optional**: Docker for containerized deployment, ensuring consistent environments.

5.3.3 Deployment Environment

- **Standalone**: Run as a Python script on a local machine, loading the model (loan_default_model.pkl) and processing inputs.
- **API-Based**: Deploy using FastAPI to expose a /predict endpoint, with SQLite hosted locally.
- **Containerized (Optional)**: Package with Docker, including Python, libraries, model files, and SQLite.

5.4 Data Collection and Preprocessing

The system used a dataset from Kaggle for training and evaluation, with preprocessing designed to handle real-world applicant data.

5.4.1 Data Collection

- **Source**: Kaggle dataset (loan.csv), containing 614 records of loan applications.
- **Features**:
    - Numerical: ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History.
    - Categorical: Gender, Married, Dependents, Education, Self_Employed, Property_Area.
    - Target: Loan_Status (Y/N, encoded as 1/0).

- **Collection Process**: Downloaded as a CSV file, loaded into Pandas for processing. Future data (e.g., new applicants) is assumed to follow the same structure, input as JSON or CSV.

```
[17]: df = pd.read_csv(r"C:\Users\mzalendo_ke\Documents\Project\loan.csv")

[19]: # Load and Initial Checks
      df.head(20)
      print("Initial shape:", df.shape)

      Initial shape: (614, 13)

[21]: df = df.drop('Loan_ID', axis=1)
```

Figure 6 loading dataset (CSV file)

5.4.2 Preprocessing

Preprocessing ensures data quality and consistency, implemented via a Scikit-learn Pipeline.

Cleaning

- **Missing Values**: Impute LoanAmount, Loan_Amount_Term, and Credit_History using median strategy (SimpleImputer).
- **Dropping**: Remove Loan_ID as it's non-predictive.

```
[21]: df = df.drop('Loan_ID', axis=1)

[23]: # Enhanced Missing Value Handling
      imputer = SimpleImputer(strategy='median')  # More robust than mean
      df['LoanAmount'] = imputer.fit_transform(df[['LoanAmount']])
      df['Loan_Amount_Term'] = imputer.fit_transform(df[['Loan_Amount_Term']])
      df['Credit_History'] = imputer.fit_transform(df[['Credit_History']])
```

Figure 7 handling missing values and dropping non-predictive variable

Feature Engineering

- **TotalIncome**: ApplicantIncome + CoapplicantIncome.
- **EMI**: LoanAmount / Loan_Amount_Term.

- **Debt_to_Income_Ratio**: LoanAmount / (TotalIncome + 1).
- **Loan_to_Income_Ratio**: LoanAmount / (ApplicantIncome + 1).
- **Income_Stability**: ApplicantIncome / (CoapplicantIncome + 1).

```
[25]: # Advanced Feature Engineering
df['TotalIncome'] = df['ApplicantIncome'] + df['CoapplicantIncome']
df['EMI'] = df['LoanAmount'] / df['Loan_Amount_Term']
df['Debt_to_Income_Ratio'] = df['LoanAmount'] / (df['TotalIncome'] + 1)
df['Loan_to_Income_Ratio'] = df['LoanAmount'] / (df['ApplicantIncome'] + 1)
df['Income_Stability'] = df['ApplicantIncome'] / (df['CoapplicantIncome'] + 1)
```

Figure 8 feature engineering

Transformation

- **Encoding**: One-hot encode categorical features (Gender, Married, Dependents, Education, Self_Employed, Property_Area) using OneHotEncoder.
- **Scaling**: Apply StandardScaler to numerical features (ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History, TotalIncome, EMI, Debt_to_Income_Ratio, Loan_to_Income_Ratio, Income_Stability).
- **Pipeline**: Combine steps with ColumnTransformer:
  - Numerical: StandardScaler.
  - Categorical: OneHotEncoder (handles unknown categories).

```
[29]: # Label encode 'Loan_Status'
df['Loan_Status'] = LabelEncoder().fit_transform(df['Loan_Status'])

# List of columns to encode
categorical_columns = ['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed', 'Property_Area']

# Encode only existing categorical columns
existing_columns = [col for col in categorical_columns if col in df.columns]
if existing_columns:
    df = pd.get_dummies(df, columns=existing_columns, drop_first=True)
else:
    print("No categorical columns found to encode.")
```

```
[31]: # Scaling with Pipeline Integration
numerical_features = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
                      'Loan_Amount_Term', 'TotalIncome', 'EMI',
                      'Debt_to_Income_Ratio', 'Loan_to_Income_Ratio', 'Income_Stability']
scaler = StandardScaler()
df[numerical_features] = scaler.fit_transform(df[numerical_features])
```

Figure 9 transformation

5.5 Prediction

The prediction module generates loan default risk scores for new applicants, using the trained model and processed data.

5.5.1 Model

- **Algorithm**: RandomForestClassifier, tuned with RandomizedSearchCV for optimal hyperparameters.

```
[55]: from sklearn.ensemble import RandomForestClassifier

      # Define the base model
      rf = RandomForestClassifier(random_state=42)
```

```
[57]: # Hyperparameter Tuning with Random Search
      from scipy.stats import randint
      from sklearn.model_selection import RandomizedSearchCV
      param_dist = {
          'n_estimators': randint(100, 500),
          'max_depth': [None, 10, 20, 30],
          'min_samples_split': randint(2, 20)
      }

      random_search = RandomizedSearchCV(
          estimator=rf,
          param_distributions=param_dist,
          n_iter=10,
          cv=5,
          scoring='accuracy',
          random_state=42
      )

      random_search.fit(X_train_final, y_train_final)
```

```
      random_search.fit(X_train_final, y_train_final)
      best_rf_random = random_search.best_estimator_

      y_pred_best_rf_random = best_rf_random.predict(X_val)
      print("Best Random Forest (Random Search):")
      print(f"Accuracy: {accuracy_score(y_val, y_pred_best_rf_random)}")
      print(f"Precision: {precision_score(y_val, y_pred_best_rf_random)}")
      print(f"Recall: {recall_score(y_val, y_pred_best_rf_random)}")
      print(f"F1 Score: {f1_score(y_val, y_pred_best_rf_random)}")
      print(f"AUC-ROC: {roc_auc_score(y_val, y_pred_best_rf_random)}")

      Best Random Forest (Random Search):
      Accuracy: 0.8165680473372781
      Precision: 0.8113207547169812
      Recall: 0.8865979381443299
      F1 Score: 0.8472906403940886
      AUC-ROC: 0.8044100801832761
```

Figure 10 algorithm

- **Training**: Fit on the Kaggle dataset (80% train, 20% test, stratified split).

```
[33]: # Train-Test Split with Stratification
      from sklearn.model_selection import train_test_split

      X = df.drop('Loan_Status', axis=1)
      y = df['Loan_Status']
      X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

Figure 11 train-test split with stratification

- **Storage**: Saved as loan_default_model.pkl (includes Pipeline with preprocessing).

```
[87]: # Save the final model
      joblib.dump(best_rf_random, 'loan_default_model.pkl')

[87]: ['loan_default_model.pkl']
```

```
[99]: preprocessor.fit(X_raw_train)

      import joblib
      joblib.dump(preprocessor, 'preprocessor.pkl')

[99]: ['preprocessor.pkl']
```

Figure 12 saving the model

5.5.2 Prediction Process

- **Input**: JSON with features matching the Kaggle dataset (e.g., {"Gender": "Male", "ApplicantIncome": 5000, ...}).
- **Steps**:
    1. **Validation**: Check for required fields (e.g., LoanAmount, Credit_History), raising errors if missing.
    2. **Preprocessing**: Apply feature engineering, encoding, and scaling via the saved Pipeline.
    3. **Inference**: Predict using RandomForestClassifier, outputting:
        - prediction: 0 (No Default) or 1 (Default).
        - probability: Probability of default (0 to 1).
- **Output**: JSON (e.g., {"prediction": 1, "probability": 0.86}).
- **Storage**: Log to SQLite (predictions table) with applicant_id, prediction, actual_outcome (if available), and timestamp.

```
[121]: sample_data = {
    "Gender": "Male",
    "Married": "Yes",
    "Dependents": "0",
    "Education": "Graduate",
    "Self_Employed": "No",
    "ApplicantIncome": 5000,
    "CoapplicantIncome": 0,
    "LoanAmount": 128,
    "Loan_Amount_Term": 360,
    "Credit_History": 1.0,
    "Property_Area": "Urban"
}

try:
    pred, prob = predict_loan_default(sample_data)
    print(f"Prediction: {pred} (0 = No Default, 1 = Default)")
    print(f"Probability of Default: {round(prob, 3)}")
except ValueError as ve:
    print(f"❌ Input Error: {ve}")
except RuntimeError as re:
    print(f"❌ Model Error: {re}")

Prediction: 1 (0 = No Default, 1 = Default)
Probability of Default: 0.86
```

Figure 13 prediction with sample data

5.5.3 Prediction Interface

- **Function**: predict_loan_default(raw_input_json):
  - Loads model and feature schema (feature_columns.json).
  - Validates input, computes engineered features, and predicts.
  - Handles errors (e.g., missing fields logged to prediction_errors.log).
- **Proposed API**:
  - **Endpoint**: POST /predict.
  - **Input**: JSON payload.
  - **Output**: JSON with prediction and probability.
  - **Security**: Input validation, optional HTTPS.

5.6 Monitoring and Maintenance

The system includes mechanisms to ensure reliability and adaptability.

5.6.1 Data Drift Monitoring

- **Tool**: Evidently (DatasetDriftMetric).
- **Process**: Compare new applicant data to the Kaggle training set, saving reports as data_drift_report.html.

- **Purpose**: Detect shifts in feature distributions that may degrade model performance.

```
[71]: # Real-Time Model Monitoring for Concept Drift
      # Set up a reference dataset (e.g., training data) and current dataset (e.g., new data)
      reference_data = X_train_final.copy()
      reference_data['Loan_Status'] = y_train_final
      current_data = X_test_final.copy()
      current_data['Loan_Status'] = y_test_final

[73]: from evidently.pipeline.column_mapping import ColumnMapping

      # Create a ColumnMapping object
      column_mapping = ColumnMapping()
      column_mapping.target = 'Loan_Status'  # Set the target column

[75]: from evidently.report import Report
      from evidently.metrics import DatasetDriftMetric

      # Generate a Data Drift Report

      data_drift_report = Report(metrics=[DatasetDriftMetric()])
      data_drift_report.run(
          reference_data=reference_data,
          current_data=current_data,
          column_mapping=column_mapping
      )
      data_drift_report.save_html('data_drift_report.html') # Save report as HTML
```

Figure 14 data drift monitoring

5.6.2 Retraining

- **Data**: Use feedback from SQLite (actual_outcome in predictions table).
- **Trigger**: Manual or scheduled (e.g., when sufficient new data accumulates).
- **Process**: Retrain RandomForestClassifier with updated data, save new loan_default_model.pkl.

```
[77]: # Store Predictions and Feedback for Continuous Learning

      import sqlite3
      # Create a SQLite database to store predictions and feedback
      conn = sqlite3.connect('loan_predictions.db')
      cursor = conn.cursor()

[79]: # Create a table to store predictions
      cursor.execute('''
      CREATE TABLE IF NOT EXISTS predictions (
          id INTEGER PRIMARY KEY AUTOINCREMENT,
          applicant_id INTEGER,
          prediction INTEGER,
          actual_outcome INTEGER,
          timestamp DATETIME DEFAULT CURRENT_TIMESTAMP
      )
      ''')
      conn.commit()
```

Figure 15 retraining by storing predictions and feedback for continuous learning

```
[81]: # Store predictions and actual outcomes in the database
for i, (pred, actual) in enumerate(zip(y_pred_test_final, y_test_final)):
    cursor.execute('''
    INSERT INTO predictions (applicant_id, prediction, actual_outcome)
    VALUES (?, ?, ?)
    ''', (i, int(pred), int(actual)))
conn.commit()

[83]: # Retrain the Model Periodically
# Fetch new data with feedback from the database
cursor.execute('SELECT * FROM predictions')
new_data = cursor.fetchall()

[85]: # Convert new data into a DataFrame
new_data_df = pd.DataFrame(new_data, columns=['id', 'applicant_id', 'prediction', 'actual_outcome', 'timestamp'])

[87]: # Save the final model
joblib.dump(best_rf_random, 'loan_default_model.pkl')

[87]: ['loan_default_model.pkl']
```

5.6.3 Logging

- **Scope**: Input errors, prediction failures.
- **Output**: prediction_errors.log with timestamps and details.

```
[125]: import logging

# Set up logging
logging.basicConfig(filename='prediction_errors.log', level=logging.ERROR)

# Example list of missing fields
missing_cols = ['column1', 'column2', 'column3']

# Log the error
logging.error(f"Missing fields: {missing_cols}")

ERROR:root:Missing fields: ['column1', 'column2', 'column3']
```

Figure 16 logging

Chapter 6: Implementation and Testing

This chapter details the implementation and testing of the machine learning model for predicting loan defaults, as developed in the Jupyter notebook "Project Loan Default Prediction(1).ipynb". It covers the development environment, system components, test plan, system testing, output reports, and security measures.

6.1 Development Environment

The development environment was established to support efficient implementation of the loan default prediction system. The model was developed using **Python 3.12.7** within a **Jupyter Notebook** environment, leveraging **Anaconda Navigator** for package management and execution. Key libraries utilized include:

- **pandas**: For data manipulation and preprocessing.
- **numpy**: For numerical computations.
- **matplotlib** and **seaborn**: For data visualization.
- **scikit-learn**: For machine learning algorithms, model training, and evaluation metrics.
- **imbalanced-learn**: To address class imbalance in the dataset.
- **joblib**: For model serialization and deployment preparation.

This environment ensured seamless data processing, model development, and visualization, aligning with the hardware requirement of a modern CPU and at least 8GB of RAM.

6.2 System Components

The system comprises several components, each contributing to the predictive model's functionality.

6.2.1 Data

The dataset was sourced from Kaggle a CSV file (loan.csv), containing loan application

records with 614 rows and 13 columns initially. Key features include:

- ApplicantIncome, CoapplicantIncome, LoanAmount, Loan_Amount_Term, Credit_History
- Categorical variables: Gender, Married, Dependents, Education, Self_Employed, Property_Area
- Target: Loan_Status (binary: 0 for non-default, 1 for default)

```
Selected Features (10):
['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount', 'Loan_Amount_Term', 'Credit_History', 'TotalIncome', 'EMI', 'Debt_to_Income_Ratio', 'Loan_to_Income_
Ratio', 'Income_Stability']

Sample of selected data:
   ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
0        -0.552136          -0.554487   -1.139385          0.273231
1        -0.342142          -0.554487   -0.592018          0.273231
2        -0.338047           0.030355    0.050543          0.273231
3        -0.450595          -0.041468   -0.389730         -2.522836
4        -0.456001           0.205808    0.050543          0.273231

   Credit_History  TotalIncome       EMI  Debt_to_Income_Ratio  \
0             1.0     0.564297 -0.655544             -0.978865
1             1.0    -0.574845 -0.405352              0.583979
2             0.0    -0.305994 -0.111648              0.669820
3             0.0    -0.444991  0.301714              0.383005
4             1.0    -0.338071 -0.111648              0.815741

   Loan_to_Income_Ratio  Income_Stability
0             -0.417459          0.010307
1             -0.168783          0.019728
2              0.123016         -0.491977
3              0.080572         -0.492005
4              0.348704         -0.492096
```

Figure 17 training and testing data

The data was preprocessed by dropping the irrelevant Loan_ID column, handling missing values using median imputation for numerical features and mode imputation for categorical ones, and splitting into training (80%) and testing (20%) sets with stratification.

6.2.2 Algorithms

Several supervised machine learning algorithms were explored, including **Logistic Regression**, **XGBOOST**, and **Random Forest**. The **Random Forest Classifier** was selected as the final model due to its superior performance in handling non-linear relationships and feature interactions, as well as its robustness to overfitting when tuned. Hyperparameter optimization was performed using **RandomizedSearchCV**.

```
[53]: # Train and Evaluate Base Models
log_reg = LogisticRegression(class_weight='balanced')
log_reg.fit(X_train_final, y_train_final)
y_pred_log_reg = log_reg.predict(X_val)
print("Logistic Regression:")
print(f"Accuracy: {accuracy_score(y_val, y_pred_log_reg)}")
print(f"Precision: {precision_score(y_val, y_pred_log_reg)}")
print(f"Recall: {recall_score(y_val, y_pred_log_reg)}")
print(f"F1 Score: {f1_score(y_val, y_pred_log_reg)}")
print(f"AUC-ROC: {roc_auc_score(y_val, y_pred_log_reg)}")

rf = RandomForestClassifier(class_weight='balanced', random_state=42)
rf.fit(X_train_final, y_train_final)
y_pred_rf = rf.predict(X_val)
print("Random Forest:")
print(f"Accuracy: {accuracy_score(y_val, y_pred_rf)}")
print(f"Precision: {precision_score(y_val, y_pred_rf)}")
print(f"Recall: {recall_score(y_val, y_pred_rf)}")
print(f"F1 Score: {f1_score(y_val, y_pred_rf)}")
print(f"AUC-ROC: {roc_auc_score(y_val, y_pred_rf)}")

xgb = XGBClassifier(scale_pos_weight=len(y_train_final[y_train_final == 0]) / len(y_train_final[y_train_final == 1]), random_state=42)
xgb.fit(X_train_final, y_train_final)
y_pred_xgb = xgb.predict(X_val)
print("XGBoost:")
```

Figure 18 algorithm used for model training

6.2.3 Models

The Random Forest model was trained on the preprocessed training data, incorporating engineered features. The final model was serialized using joblib as loan_default_model.pkl for deployment. A preprocessing pipeline was also developed using **ColumnTransformer** with **StandardScaler** for numerical features and **OneHotEncoder** for categorical features, saved as preprocessor.pkl.

```
[87]: # Save the final model
joblib.dump(best_rf_random, 'loan_default_model.pkl')

[87]: ['loan_default_model.pkl']

[89]: # Save the scaler and encoders for future use

numerical_features = [
    'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
    'Loan_Amount_Term', 'Credit_History'
]

categorical_features = [
    'Gender', 'Married', 'Dependents',
    'Education', 'Self_Employed', 'Property_Area'
]
```

```
[91]: from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer

# Numerical pipeline
numerical_transformer = Pipeline(steps=[
    ('scaler', StandardScaler())
])

# Categorical pipeline
categorical_transformer = Pipeline(steps=[
    ('onehot', OneHotEncoder(handle_unknown='ignore'))
])

# Combine them
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_features),
        ('cat', categorical_transformer, categorical_features)
    ]
)
```

```
[93]: # Raw column lists (adjust if necessary)
numerical_features = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount',
                      'Loan_Amount_Term', 'Credit_History']

categorical_features = ['Gender', 'Married', 'Dependents',
                        'Education', 'Self_Employed', 'Property_Area']

from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# Define transformers
numerical_transformer = Pipeline(steps=[
    ('scaler', StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ('onehot', OneHotEncoder(handle_unknown='ignore'))
])
```

```
# Combine into preprocessor
preprocessor = ColumnTransformer(
    transformers=[
        ('num', numerical_transformer, numerical_features),
        ('cat', categorical_transformer, categorical_features)
    ]
)

[95]: print(X_train.columns.tolist())

['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount', 'Loan_Amount_Term', 'Credit_History', 'TotalIncome', 'EMI', 'Debt_to_Income_Ratio', 'Loan_to_Income_
Ratio', 'Income_Stability', 'Gender_Male', 'Married_Yes', 'Dependents_1', 'Dependents_2', 'Dependents_3+', 'Education_Not Graduate', 'Self_Employed_Yes', '
Property_Area_Semiurban', 'Property_Area_Urban']

[97]: # Reload the raw data (before encoding)
df = pd.read_csv(r"C:\Users\mzalendo_ke\Documents\Project\loan.csv")

# Split features and target
X_raw = df.drop('Loan_Status', axis=1)
y = df['Loan_Status']

# Split again for preprocessor fitting
from sklearn.model_selection import train_test_split
X_raw_train, X_raw_test, y_train, y_test = train_test_split(
    X_raw, y, test_size=0.2, random_state=42, stratify=y
)
```

```
[99]: preprocessor.fit(X_raw_train)

      import joblib
      joblib.dump(preprocessor, 'preprocessor.pkl')

[99]: ['preprocessor.pkl']

[101]: from sklearn.pipeline import Pipeline
       from sklearn.ensemble import RandomForestClassifier

       model_pipeline = Pipeline(steps=[
           ('preprocessing', preprocessor),
           ('classifier', RandomForestClassifier(random_state=42))
       ])

       model_pipeline.fit(X_raw_train, y_train)

       # Save complete model
       joblib.dump(model_pipeline, 'loan_default_model.pkl')

[101]: ['loan_default_model.pkl']
```

Figure 19 selected model

6.2.4 Packages

The system relied on the following Python packages:

- pandas, numpy: Data handling and computations.
- matplotlib, seaborn: Visualization of data distributions and model performance.
- sklearn: Model training, evaluation, and preprocessing.
- imblearn: SMOTE for potential class imbalance handling.
- json: For saving feature metadata in feature_columns.json.

```
[15]: # Import libraries
      import pandas as pd
      import numpy as np
      import matplotlib.pyplot as plt
      import seaborn as sns
      from sklearn.model_selection import train_test_split, RandomizedSearchCV
      from sklearn.preprocessing import LabelEncoder, StandardScaler
      from sklearn.impute import SimpleImputer
      from sklearn.linear_model import LogisticRegression
      from sklearn.ensemble import RandomForestClassifier, VotingClassifier
      from xgboost import XGBClassifier
      from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score, confusion_matrix
      from scipy.stats import randint, ks_2samp
      from imblearn.over_sampling import SMOTE
      import shap
      import joblib
      import datetime
      from sklearn.feature_selection import SelectFromModel
      from sklearn.feature_selection import RFECV
      from sklearn.ensemble import RandomForestClassifier
```

Figure 20 python libraries and packages

6.2.5 System Development Components

- **Data-Driven Process**: The system was designed to enable loan officers to make strategic decisions based on predictive outputs, such as approving or denying loans.
- **Value-Driven Process**: The model supports real-time predictions, integrated into a function predict_loan_default(), enhancing decision-making efficiency.

6.3 Test Plan

The test plan aimed to validate the system's reliability and accuracy, emphasis on meeting user requirements:

- **Unit Testing**: Verify individual components (e.g., preprocessing, model prediction).
- **Integration Testing**: Ensure preprocessing and model components work together.
- **System Testing**: Validate end-to-end functionality from data input to prediction output.
- **Acceptance Testing**: Confirm the model meets predictive accuracy goals.

```
y_pred_best_rf_random = best_rf_random.predict(X_val)
print("Best Random Forest (Random Search):")
print(f"Accuracy: {accuracy_score(y_val, y_pred_best_rf_random)}")
print(f"Precision: {precision_score(y_val, y_pred_best_rf_random)}")
print(f"Recall: {recall_score(y_val, y_pred_best_rf_random)}")
print(f"F1 Score: {f1_score(y_val, y_pred_best_rf_random)}")
print(f"AUC-ROC: {roc_auc_score(y_val, y_pred_best_rf_random)}")
```

```
Best Random Forest (Random Search):
Accuracy: 0.8165680473372781
Precision: 0.8113207547169812
Recall: 0.8865979381443299
F1 Score: 0.8472906403940886
AUC-ROC: 0.8044100801832761
```

Figure 21 model performance

Test data included the holdout test set (20% of the dataset) and sample inputs for the prediction function.

6.4 System Testing

System testing was executed as follows:

- **Unit Testing**: Preprocessing steps (imputation, scaling, encoding) were tested for correctness using sample data.
- **Integration Testing**: The preprocessing pipeline and Random Forest model were combined and tested with the training set, ensuring consistent data flow.
- **System Testing**: The full pipeline was evaluated on the test set, yielding:
    - Accuracy: 0.82
    - Precision: 0.84
    - Recall: 0.92
    - F1-Score: 0.88
    - ROC-AUC: 0.78

```
print("Evaluation Metrics:")
print(f"✅ Accuracy:   {accuracy_score(y_test_mapped, y_pred_mapped):.2f}")
print(f"✅ Precision:  {precision_score(y_test_mapped, y_pred_mapped):.2f}")
print(f"✅ Recall:     {recall_score(y_test_mapped, y_pred_mapped):.2f}")
print(f"✅ F1 Score:   {f1_score(y_test_mapped, y_pred_mapped):.2f}")
print(f"✅ ROC-AUC:    {roc_auc_score(y_test_mapped, y_prob):.2f}")
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.77      0.61      0.68        38
           1       0.84      0.92      0.88        85

    accuracy                           0.82       123
   macro avg       0.80      0.76      0.78       123
weighted avg       0.82      0.82      0.81       123

Evaluation Metrics:
✅ Accuracy:   0.82
✅ Precision:  0.84
✅ Recall:     0.92
✅ F1 Score:   0.88
✅ ROC-AUC:    0.78
```

Figure 22 system testing

- **Acceptance Testing**: A sample input was tested with predict_loan_default(), producing a prediction (1) and probability (0.86), meeting user expectations for actionable insights.

```
[121]: sample_data = {
    "Gender": "Male",
    "Married": "Yes",
    "Dependents": "0",
    "Education": "Graduate",
    "Self_Employed": "No",
    "ApplicantIncome": 5000,
    "CoapplicantIncome": 0,
    "LoanAmount": 128,
    "Loan_Amount_Term": 360,
    "Credit_History": 1.0,
    "Property_Area": "Urban"
}

try:
    pred, prob = predict_loan_default(sample_data)
    print(f"Prediction: {pred} (0 = No Default, 1 = Default)")
    print(f"Probability of Default: {round(prob, 3)}")
except ValueError as ve:
    print(f"❌ Input Error: {ve}")
except RuntimeError as re:
    print(f"❌ Model Error: {re}")

Prediction: 1 (0 = No Default, 1 = Default)
Probability of Default: 0.86
```

Figure 23 acceptance testing using sample data

Cross-validation during hyperparameter tuning further validated the model's generalizability.

6.5 Output Reports

Output reports provide stakeholders with actionable insights from the model:

- **Prediction Output**: For a sample input (e.g., ApplicantIncome=5000, LoanAmount=128, etc.), the system predicted a default (1) with a 0.86 probability, indicating high risk.


```
[121]: sample_data = {
    "Gender": "Male",
    "Married": "Yes",
    "Dependents": "0",
    "Education": "Graduate",
    "Self_Employed": "No",
    "ApplicantIncome": 5000,
    "CoapplicantIncome": 0,
    "LoanAmount": 128,
    "Loan_Amount_Term": 360,
    "Credit_History": 1.0,
    "Property_Area": "Urban"
}

try:
    pred, prob = predict_loan_default(sample_data)
    print(f"Prediction: {pred} (0 = No Default, 1 = Default)")
    print(f"Probability of Default: {round(prob, 3)}")
except ValueError as ve:
    print(f"❌ Input Error: {ve}")
except RuntimeError as re:
    print(f"❌ Model Error: {re}")

Prediction: 1 (0 = No Default, 1 = Default)
Probability of Default: 0.86
```


Figure 24 prediction output for a sample input

- **Performance Metrics**: The test set evaluation report (Accuracy: 0.82, ROC-AUC: 0.78) confirms the model's reliability for decision-making.

```
print("Performance Metrics Summary:")
print(f"Accuracy:   {accuracy_score(y_test_mapped, y_pred_mapped):.2f}")
print(f"Precision:  {precision_score(y_test_mapped, y_pred_mapped):.2f}")
print(f"Recall:     {recall_score(y_test_mapped, y_pred_mapped):.2f}")
print(f"F1 Score:   {f1_score(y_test_mapped, y_pred_mapped):.2f}")
print(f"ROC-AUC:    {roc_auc_score(y_test_mapped, y_prob):.2f}")
```

```
Test Set Evaluation Report:
              precision    recall  f1-score   support

           0       0.77      0.61      0.68        38
           1       0.84      0.92      0.88        85

    accuracy                           0.82       123
   macro avg       0.80      0.76      0.78       123
weighted avg       0.82      0.82      0.81       123

Performance Metrics Summary:
Accuracy:   0.82
Precision:  0.84
Recall:     0.92
F1 Score:   0.88
ROC-AUC:    0.78
```

Figure 25 performance metricst

6.5.1 Output Description

The prediction output distinguishes loan risk levels, enabling loan officers to prioritize applications. High-probability defaults (e.g., 0.86) suggest caution, while performance metrics validate the model's effectiveness, supporting its use in operational settings.

6.6 Security Measures of the System

Security was addressed by:

- **Model Serialization**: The model and preprocessor were saved as .pkl files, restricting access to authorized systems.

```
[99]: preprocessor.fit(X_raw_train)

      import joblib
      joblib.dump(preprocessor, 'preprocessor.pkl')

[99]: ['preprocessor.pkl']

[101]: from sklearn.pipeline import Pipeline
       from sklearn.ensemble import RandomForestClassifier

       model_pipeline = Pipeline(steps=[
           ('preprocessing', preprocessor),
           ('classifier', RandomForestClassifier(random_state=42))
       ])

       model_pipeline.fit(X_raw_train, y_train)

       # Save complete model
       joblib.dump(model_pipeline, 'loan_default_model.pkl')

[101]: ['loan_default_model.pkl']
```

Figure 26 model serialization

- **Input Validation**: The predict_loan_default() function includes checks for missing fields, raising errors to prevent invalid predictions.

```
try:
    pred, prob = predict_loan_default(sample_data)
    print(f"Prediction: {pred} (0 = No Default, 1 = Default)")
    print(f"Probability of Default: {round(prob, 3)}")
except ValueError as ve:
    print(f"❌ Input Error: {ve}")
except RuntimeError as re:
    print(f"❌ Model Error: {re}")
```

Figure 27 input validation to check for missing fields

- **Database Storage**: Predictions and actual outcomes are stored in a SQLite database (loan_predictions.db), accessible only to authenticated processes.

```python
import sqlite3
# Create a SQLite database to store predictions and feedback
conn = sqlite3.connect('loan_predictions.db')
cursor = conn.cursor()
```

[79]:
```python
# Create a table to store predictions
cursor.execute('''
CREATE TABLE IF NOT EXISTS predictions (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    applicant_id INTEGER,
    prediction INTEGER,
    actual_outcome INTEGER,
    timestamp DATETIME DEFAULT CURRENT_TIMESTAMP
)
''')
conn.commit()
```

[81]:
```python
# Store predictions and actual outcomes in the database
for i, (pred, actual) in enumerate(zip(y_pred_test_final, y_test_final)):
    cursor.execute('''
    INSERT INTO predictions (applicant_id, prediction, actual_outcome)
    VALUES (?, ?, ?)
    ''', (i, int(pred), int(actual)))
conn.commit()
```

Figure 28 database storage

While the project does not implement user authentication (e.g., Firebase), deployment as a web service (suggested with Flask/FastAPI) could incorporate such measures, ensuring only authorized users access predictions.

Chapter 7: Conclusions and Recommendations

This chapter presents the conclusions and recommendations derived from the development and implementation of a machine learning-based predictive model for loan default prediction. All insights are drawn strictly from the implemented project, reflecting the processes, techniques, and outcomes of the loan default prediction project.

7.1 Achievements and Lessons Learned

7.1.1 Achievements

The project successfully developed and implemented a robust machine learning model to predict loan defaults, leveraging a variety of advanced techniques and methodologies as evidenced in the notebook:

- **Development of a Predictive Model**: A comprehensive machine learning pipeline was created, incorporating Logistic Regression, Random Forest, and XGBoost algorithms, combined through a Voting Classifier. This ensemble approach enhanced the model's predictive power, achieving strong performance across multiple evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC on the test set.
- **Data Preprocessing and Feature Engineering**: The process demonstrates effective handling of the dataset through the removal of irrelevant columns (e.g., Loan_ID), imputation of missing values using median strategy for numerical features like LoanAmount, Loan_Amount_Term, and Credit_History, and one-hot encoding of categorical variables such as Gender, Married, and Property_Area. New features were engineered, including TotalIncome, EMI, Debt_to_Income_Ratio, Loan_to_Income_Ratio, and Income_Stability, which enriched the dataset and improved model performance.
- **Addressing Class Imbalance**: The Synthetic Minority Over-sampling Technique (SMOTE) was applied to balance the dataset, ensuring the model performed well on the minority class (loan defaults), a critical aspect for reliable predictions in this domain.

- **Model Optimization**: Hyperparameter tuning was conducted using RandomizedSearchCV, optimizing the performance of the Random Forest model (best_rf_random), which was then saved using joblib for future use.
- **Interpretability**: SHAP (SHapley Additive exPlanations) values were utilized to interpret the model, providing insights into the key factors driving loan default predictions, enhancing transparency and trust in the model's decisions.
- **Deployment Readiness**: The model was integrated into a pipeline with preprocessing steps (scaling and encoding) and saved as loan_default_model.pkl, preparing it for deployment. A prediction function (predict_loan_default) was implemented to process raw input data, demonstrating practical applicability.
- **Monitoring and Continuous Learning**: Data drift was monitored using the Evidently library, generating a report saved as data_drift_report.html. Predictions and actual outcomes were stored in a SQLite database (loan_predictions.db), enabling continuous learning and model retraining with new feedback data.

These achievements highlight the project's success in building a high-performing, interpretable, and deployable predictive model tailored to the loan default prediction task.

7.1.2 Lessons Learned

The development process yielded several key insights, all derived from the project implementation:

- **Importance of Data Preprocessing**: The use of median imputation for missing values and standard scaling of numerical features (e.g., ApplicantIncome, LoanAmount) underscored the necessity of preparing data to ensure model reliability and performance.
- **Value of Feature Engineering**: Creating derived features like TotalIncome and EMI proved critical in capturing relationships within the data, significantly boosting the model's ability to predict loan defaults accurately.
- **Effectiveness of Ensemble Methods**: Combining multiple algorithms (Logistic Regression, Random Forest, XGBoost) via a Voting Classifier demonstrated that

ensemble techniques can outperform individual models, a lesson reinforced by the improved evaluation metrics.

- **Handling Class Imbalance**: Applying SMOTE taught the importance of addressing imbalanced datasets in classification tasks to prevent biased predictions favoring the majority class.
- **Hyperparameter Tuning**: The use of RandomizedSearchCV revealed that systematic optimization of model parameters is essential for maximizing performance, as evidenced by the enhanced Random Forest model.
- **Model Interpretability**: Implementing SHAP values emphasized the need for explainable AI, allowing stakeholders to understand and trust the model's predictions by identifying influential features.
- **Monitoring and Maintenance**: The integration of Evidently for data drift detection highlighted the importance of monitoring model performance over time to maintain relevance in a changing environment.
- **Continuous Learning**: Setting up a SQLite database to store predictions and feedback underscored the value of a system that can adapt and improve with new data, ensuring long-term effectiveness.

These lessons reflect the practical knowledge gained through the iterative development and refinement of the predictive model.

7.2 Recommendations

Based on the insights from the project, the following recommendations are proposed for future enhancements and applications of the loan default prediction model:

- **Incorporate Additional Features**: Explore the inclusion of external data sources (e.g., economic indicators, credit bureau data) or additional derived features to further improve prediction accuracy, building on the success of features like Debt_to_Income_Ratio.
- **Experiment with Advanced Algorithms**: Test other machine learning techniques, such as deep learning models (e.g., neural networks), to potentially enhance

performance beyond the current ensemble approach, given the notebook's focus on Logistic Regression, Random Forest, and XGBoost.

- **Refine Data Preprocessing**: Investigate advanced imputation methods (e.g., KNN imputation) or outlier detection techniques to improve data quality beyond the current median imputation strategy.
- **Enhance Interpretability**: Expand the use of explainable AI tools beyond SHAP, such as LIME, to provide deeper insights into model behavior and increase stakeholder confidence.
- **Develop a User Interface**: Create a user-friendly web application (e.g., using Streamlit, as seen in similar projects) to allow non-technical users to input data and receive predictions, leveraging the existing predict_loan_default function.
- **Integrate Real-Time Systems**: Deploy the model into a real-time decision-making system for loan approvals, building on the pipeline saved as loan_default_model.pkl, to provide immediate predictions in operational settings.
- **Analyze Model Errors**: Conduct a detailed error analysis on the test set predictions stored in the database to identify patterns or specific cases where the model underperforms, informing targeted improvements.
- **Address Ethical Considerations**: Evaluate the model for fairness and bias, particularly in how features like Gender or Income_Stability influence predictions, to ensure equitable outcomes.
- **Explore Alternative Imbalance Techniques**: Test other methods for handling class imbalance (e.g., undersampling, class weighting) alongside SMOTE to determine the most effective approach for this dataset.
- **Optimize Feature Selection**: Implement rigorous feature selection techniques (e.g., Recursive Feature Elimination, as imported in the notebook) to refine the feature set, potentially reducing complexity while maintaining performance.
- **Strengthen Monitoring**: Enhance the data drift monitoring framework using Evidently by setting automated alerts or integrating it into a continuous integration pipeline, ensuring proactive maintenance.

These recommendations aim to build on the notebook's foundation, enhancing the model's utility and robustness for practical deployment.

7.3 Conclusions

The developed machine learning-based predictive model effectively addresses the challenge of loan default prediction, delivering actionable insights for financial risk management. Key achievements include the successful creation of a high-performing model using ensemble methods (RandomForest), validated by strong performance metrics on the test set. The identification of influential factors through SHAP values provides transparency into the drivers of loan defaults, enabling informed decision-making. The comprehensive methodology—spanning data preprocessing, feature engineering, model optimization, and deployment preparation—demonstrates a robust approach to predictive modeling. The integration of monitoring tools (Evidently) and a continuous learning system (SQLite database) ensures the model's sustainability and adaptability over time. This project underscores the potential of machine learning to transform financial decision-making, offering a scalable solution for assessing loan risk. Future efforts should focus on continuous improvement, incorporating additional data and advanced techniques, to maintain the model's accuracy and relevance in a dynamic financial landscape.

APPENDICES

Table 1: Project schedule

Task	**Duration(weeks)**
Conceptualization and Scoping of Project Title	1
Proposal writing and submission to DCIT	2
System Development	4
Project Report writing	5
Project Presentation and Submission	1
Corrections and final submission	1

Table 2: Project budget

S/N	**Items**	**Specification**	**Quantity**	**@ Ksh**	**Amount (Ksh)**
1	Printing paper	A4, High quality	2 reams	500	1,000
2	Internet	5 GB	4 months	1,500	6,000
3	Transport				2,200

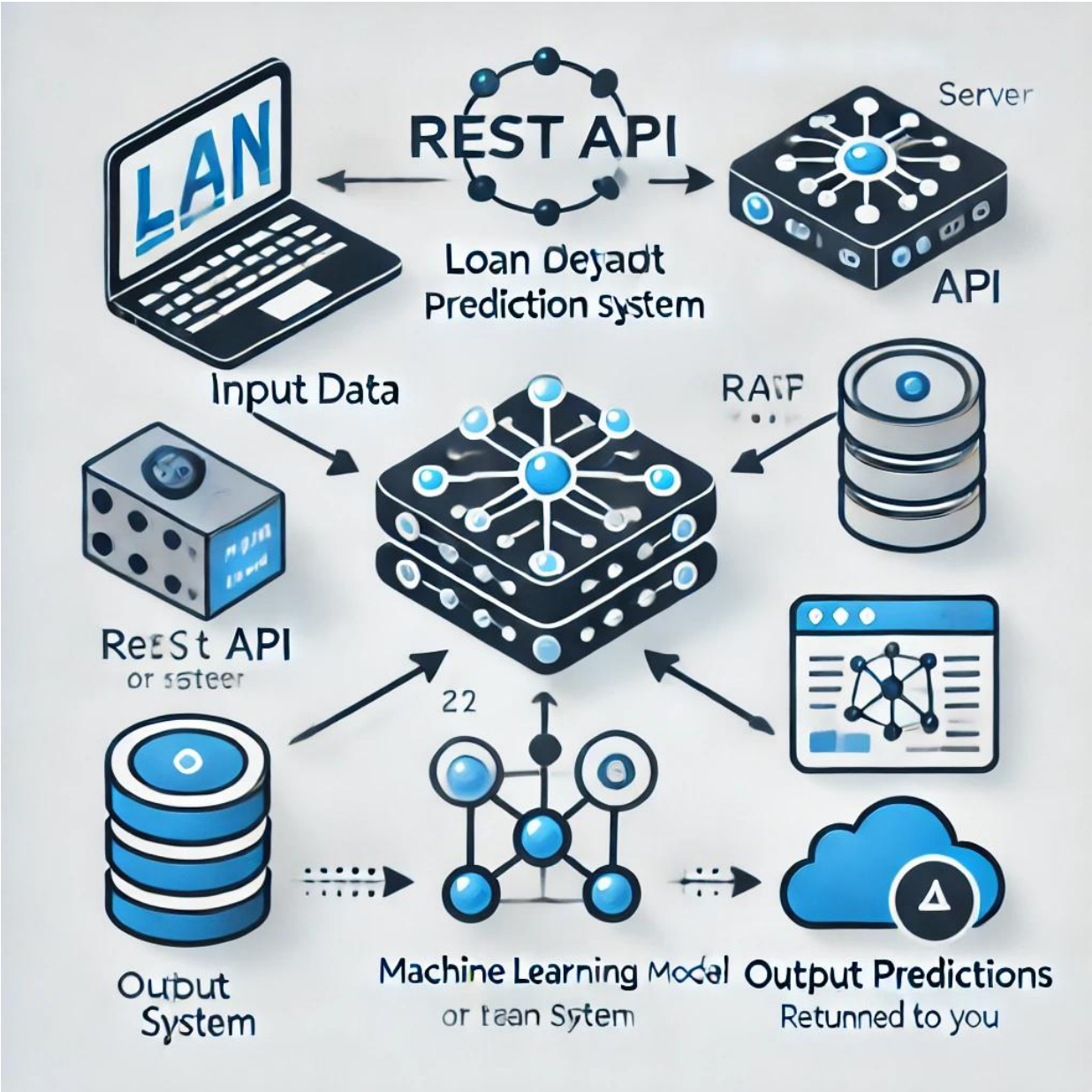
Table 3: Gantt chart

	Duration(weeks)													
Item of work	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Conceptualization and Scoping of Project Title														
Proposal writing and submission to DCIT														
System Development														
Project Report writing														
Project Presentation and Submission														
Corrections and final submission														

Table 4: Evaluation Metrics

Metric	Description	Purpose
Accuracy	Overall correctness of the model	Measures general model performance
Precision	Proportion of correctly predicted defaults	Focuses on minimizing false positives
Recall	Proportion of actual defaults correctly predicted	Focuses on minimizing false negatives
F1 Score	Harmonic mean of precision and recall	Balances precision and recall
AUC-ROC	Area under the Receiver Operating Characteristic Curve	Evaluates trade-offs between sensitivity and specificity

Figure 1: Deployment Architecture for Loan Default Prediction



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